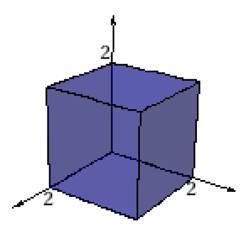
8.

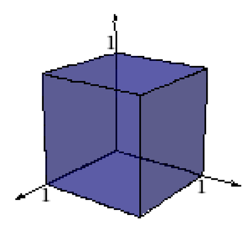
Exercício 8.1

a)



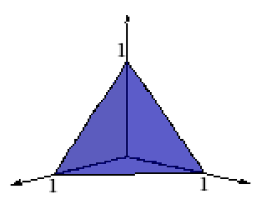
24

b)



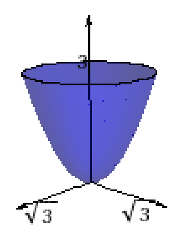
$\frac{1}{2}(-1+e)^2$

c)



$\frac{1}{120}$

d)



0

Exercício 8.2

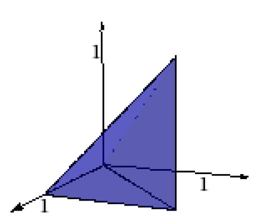
$$\int_{-\sqrt{a}}^{\sqrt{a}}\int_{-\sqrt{a-x^2}}^{\sqrt{a-x^2}}\int_{0}^{a-x^2-y^2} dz\,dy\,dx = \frac{a^2\pi}{2}$$

Exercício 8.3

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{-2\sqrt{4-x^2-y^2}}^{2\sqrt{4-x^2-y^2}} dz\,dy\,dx = \frac{64\pi}{3}$$

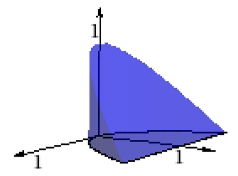
Exercício 8.4

a)



$$\int_0^1\int_z^1\int_y^1 f[x,y,z]\,dx\,dy\,dz$$

b)



$$\int_0^1 \int_0^{1-z} \int_{-\sqrt{y}}^{\sqrt{y}} f[x, y, x] \, dx \, dy \, dz$$

Exercício 8.5

1

Exercício 8.6

$\frac{\pi}{3}$

Exercício 8.7

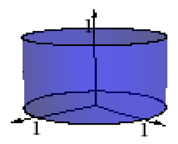
a)

$\frac{\pi}{4}$

b)

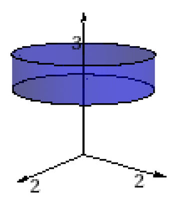
$\left(\text{Cos}[1] - \text{Cos}\left[\frac{4+\pi}{4}\right]\right) \text{Log}[2] \, \text{Sin}[1]$

Exercício 8.8



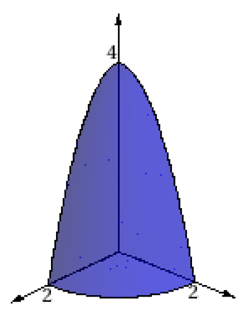
$$\int_0^1 \int_0^{2\pi} \int_0^1 z \rho^3 \, dz \, d\theta \, d\rho = \frac{\pi}{4}$$

Exercício 8.9



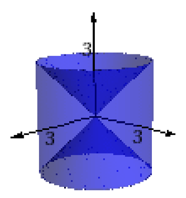
$$\int_0^2 \int_0^{2\pi} \int_2^3 z \rho \, \text{Exp}\left[\rho^2\right] dz \, d\theta \, d\rho = \frac{5}{2} \left(-1 + e^4\right) \pi$$

Exercício 8.10



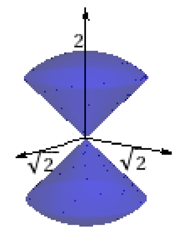
$$\int_0^{\frac{\pi}{2}} \int_0^2 \int_0^{4-\rho^2} \left(\rho^2 \, \text{Cos}[\theta] + \rho^2 \, \text{Sin}[\theta]\right) dz \, d\rho \, d\theta = \frac{128}{15}$$

Exercício 8.11



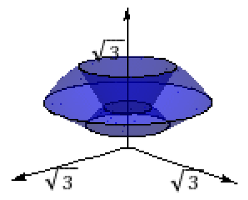
$$\int_0^{2\pi} \int_0^3 \int_{-\rho}^{\rho} \rho \, dz \, d\rho \, d\theta = 36\pi$$

Exercício 8.12



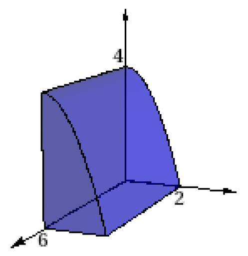
$$\int_0^{\frac{\pi}{4}}\int_0^2\int_0^{2\pi} r^2\ \mathrm{Sin}[\phi]\,d\theta\,dr\,d\phi + \int_{\frac{3\pi}{4}}^{\pi}\int_0^2\int_0^{2\pi} r^2\ \mathrm{Sin}[\phi]\,d\theta\,dr\,d\phi = 2\int_0^{\frac{\pi}{4}}\int_0^2\int_0^{2\pi} r^2\ \mathrm{Sin}[\phi]\,d\theta\,dr\,d\phi = -\frac{16}{3}\left(-2+\sqrt{2}\right)\pi$$

Exercício 8.13



$$\int_1^2\int_0^{2\pi}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \mathrm{Exp}\left[r^3\right] r^2\ \mathrm{Sin}[\phi]\ d\phi\,d\theta\,dr = \frac{1}{3}\left(-1+\sqrt{3}\right) e \left(-1+e^7\right)\pi$$

Exercício 8.14



$$\int_0^6\int_0^2\int_0^{4-y^2} 1\,dz\,dy\,dx = 32$$

Exercício 8.15

a)

$$\int_0^{2\pi}\int_0^{\pi}\int_{\sqrt{3}}^{3} r^2\ \mathrm{Sin}[\phi]\ dr\,d\phi\,d\theta = -4\left(-9+\sqrt{3}\right)\pi$$

b)

$$\int_0^{2\pi}\int_0^{\sqrt{6}}\int_{\rho^2}^{12-\rho^2} \rho\,dz\,d\rho\,d\theta = 36\pi$$

c)

$$\int_0^{2\pi}\int_0^2\int_{\frac{\rho^2}{4}}^{1} \rho\,dz\,d\rho\,d\theta = 2\pi$$

d)

$$\int_{-2}^{2}\int_{-\sqrt{9-\frac{9z^2}{4}}}^{\sqrt{9-\frac{9z^2}{4}}}\int_{y^2+\frac{9z^2}{4}}^{9} 1\,dx\,dy\,dz = 27\pi$$

e)

$$\int_0^{2\pi}\int_1^2\int_0^{\rho^2} \rho\,dz\,d\rho\,d\theta = \frac{15\pi}{2}$$

Exercício 8.16

$$2\int_0^{2\pi}\int_0^{\pi}\int_0^{R}\sqrt{R^2-r^2}\ r^2\ \mathrm{Sin}[\phi]\ dr\,d\phi\,d\theta = \frac{\pi^2 R^4}{2}$$

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